

医药翻译PRD.pdf - 纯译文

第 1 段

Medical Translation PRD

Prototype reference document is continuously being updated at <http://10.3.75.127:3000/>

I. Project Background

1.1 Business Background

The pharmaceutical industry frequently deals with multilingual document processing in daily R&D, regulatory submissions, academic exchanges, and international collaborations. Typical scenarios include:

- Bilingual translation of clinical trial reports, investigator brochures, and informed consent forms;
- Creation of multilingual versions for medical device technical documents, instructions for use, and labels;
- Rapid translation and side-by-side review of pharmaceutical academic papers and regulatory submissions (e.g., FDA, EMA, NMPA).

Currently, users mainly rely on general-purpose translation tools or manual translation. General tools lack specialized pharmaceutical terminology support, resulting in inaccurate and inconsistent translations of key terms. Manual translation is costly and time-consuming, failing to meet the demand for frequent, small-batch, real-time translations. Moreover, existing tools poorly preserve formatting in Word and PDF documents—requiring extensive manual reformatting after translation, severely impacting efficiency.

1.2 User Pain Points

Pain Point	Description
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Inaccurate Terminology	General translation engines fail to recognize pharmaceutical-specific terms such as "randomized controlled trial", "primary endpoint", and "adverse event", rendering the output unusable.
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Formatting Easily Broken	After translating PDF or Word files, tables, headers/footers, and figure layouts become disordered, requiring significant manual correction.
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Lack of Bilingual Comparison	Only translated text is provided without a side-by-side comparison to the original, necessitating manual assembly of bilingual versions.
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No Historical Traceability	Each translation requires re-uploading and language settings; previous translations cannot be reviewed or retrieved.
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第 2 段

【中文 -> 英文】

多语言配置繁琐 源语言、目标语言每次手动选择，效率低下

数据安全顾虑 医药文档涉及敏感信息，使用公共翻译工具存在数据泄露风险

多文件处理不便 一次性上传多个文档后，缺乏统一管理和对比预览能力

1.3 技术可行性

- 机器翻译 + 医药术语库的混合模式已在行业内成熟应用，可保证术语准确率；
- 文档格式保持技术（如 Apache POI、Aspose、PDF 解析引擎）可实现 Word/PDF 的版面稳定输出；
- 单点登录与数据隔离方案（SSO + 用户空间）可满足企业级安全要求。

二、产品价值

2.1 用户价值

价值点 说明

术语准确统一 内置医药通用术语库，确保同一术语全文一致、行业标准规范，可直接用于正式文档

格式零调整 翻译后 Word/PDF 保持原排版（允许长度微调，方向、结构不变），无需人工修复

效率大幅提升 文本翻译即时响应，文件翻译批量处理，历史记录一键复用，平均翻译时间缩短

80% 以上

对照审核便捷 文件翻译自动生成左右对照预览，并提供纯译文版和双语对照版下载，满足不同场景需求

历史可追溯 所有翻译任务自动保存，支持搜索、筛选、快速复现，避免重复工作

数据安全隔离 单点登录 + 用户数据隔离，保障医药文档的保密性和合规性

当前为本地开发占位译文。接入通义千问配置后可替换为真实模型翻译结果。

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Intuitive operation — Task-oriented interface, clear at a glance, no training required to get started.

2.2 Business Value

Value Point | Description

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Reduce translation costs | Machine translation replaces most manual translation, reducing outsourcing expenses; estimated annual cost reduction of 60% – 70%.

Accelerate business processes | Real-time translation capability shortens processing time for drug registration documents and clinical trial documentation from days to minutes.

Enhance international competitiveness | Rapid generation of high-quality multilingual documents supports enterprise expansion into overseas markets.

Knowledge asset accumulation | Historical records form an enterprise-specific medical translation corpus, which can improve terminology databases and support future AI training.

3. Requirements

3.1 User Authentication and Data Isolation

- As a user, I want the system to prioritize Single Sign-On (e.g., Feishu) to leverage unified corporate identity authentication, eliminating the need to remember multiple account credentials.

Acceptance Criteria:

1. The system login page should prominently display a "Feishu Login" button; clicking it redirects the user to the Feishu authorization page.

2. If Feishu SSO is unavailable due to configuration issues, the system must provide a backup username/password login form.

3. After successful login via either method, the user enters the main system interface with an active session that persists until manually logged out or timed out.

- As a user, I want the system to implement data isolation so that each user can only view and operate their own translation records, ensuring data security and privacy.

Acceptance Criteria:

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1. After User A logs in, the left-side history list displays only User A's translation tasks, with no records from other users shown.

2. User A cannot access or download User B's translation results by directly entering a URL or any other method.

3. The system must perform backend validation based on the logged-in user 's identity for all data operations to ensure data isolation security.

3.2 General Language Configuration

- As a user, I want the system to support at least bidirectional translation between Chinese and English, and as many common languages as possible (e.g., Japanese, Korean, German, French), to handle multilingual translation scenarios.

Acceptance Criteria:

1. Above the input area in both text translation and file translation modes, there should be two dropdown menus for source language and target language, allowing users to freely select any supported language.

2. The source language dropdown must include options “ Auto-detect, ” “ Chinese, ” and “ English, ” with “ Auto-detect ” as the default. When the user inputs Chinese text, the system should automatically recognize it as “ Chinese ” ;

when inputting English text, it should automatically recognize it as “ English, ” and display the recognized result in the dropdown. The same applies to file uploads.

3. If the user manually changes the source language to a specific language (e.g., “ English ”), the system must use the user-selected language for translation and no longer override it automatically.

4. The target language dropdown does not support auto-detection; the user must manually select the target language. The enumerated values must include at least “ Chinese ” and “ English. ”

3.3 Medical Industry General Terminology Database Mounting

- As a user, I want the system to call the medical industry general terminology database during translation to ensure accurate and consistent professional term translations.

Acceptance Criteria:

1. Upon submission of a translation request (text or file), the system backend must invoke the medical industry general terminology database for term matching and replacement.

2. For medical terms included in the terminology database (e.g., “ randomized controlled trial ”), the translated output must uniformly use the standard translation from the database (e.g., “ randomized controlled trial ”)—no non-database literal translations are allowed.

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3. If a term is not found in the terminology database, the system shall use a general machine translation engine to process it normally, without interrupting the translation due to missing terms.

3.4 General Translation Requirements

As a user, I want to see a translation status prompt (e.g., "Loading", "Success", or "Failure") after clicking "Send", so that I can understand the translation progress. When a failure occurs, I want to be able to retry.

Acceptance Criteria:

1. After the user clicks "Send", the button should immediately become grayed out, navigate to the result page, and display either "Translating..." or a loading animation to prevent repeated clicks.

2. Upon successful translation completion, the original text should be displayed on the left side of the page, the translated text on the right side, and a "Translation Successful" message should appear (which may briefly show and then automatically disappear).

3. If the translation fails due to network or server issues, an explicit "Translation Failed" message should be shown, along with a "Retry" button in the translation area. When the user clicks "Retry", the system should re-initiate the translation request.

4. After a successful retry, the status should update to "Success"; if the retry still fails, the "Retry" button should be redisplayed again—there is no limit on the number of retries.

3.3.1 Text Translation

As a user, I want to input medical text in text translation mode, so that I can quickly complete short texts or segments' translation tasks.

Acceptance Criteria:

1. When the user is in text translation mode, a multi-line text input box should appear on the right side, allowing the user to enter medical text up to a maximum of 5000 characters.

2. After the user inputs the text and clicks the "Send" button, the system should immediately begin processing the translation.

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3. During the translation process, the "Send" button should immediately be disabled and display "Translating..." or a loading animation until the translation is complete (whether successful or failed), at which point the button becomes usable again.

As a user, I want the translation result to appear directly on the right side of the page (not in a dialogue bubble) so that I can clearly view both the original text and the translated text, and copy the translation content.

Acceptance Criteria:

1. After text translation is completed, no dialogue bubble style should appear on the right side of the page; instead, clear cards or areas should separately display "Original Text" and "Translated Text."

- 2. The original and translated texts must be clearly labeled (e.g., with headings "Original Text" and "Translated Text") to facilitate comparison.
 - 3. Next to the translated text area, there should be a "Copy" button or icon.
 - 4. When the user clicks the copy button, the translated text should be copied to the system clipboard, and a brief prompt saying "Copied" should appear.
 - 5. The copied content must be complete, formatted as plain text, and contain no extra spaces or line breaks.
- As a user, I want text translation to support only single-turn queries without maintaining multi-turn conversation context, so that I can focus on individual translation tasks without interference.

Acceptance Criteria:

- 1. After completing one text translation, if the user enters new text and clicks send again, the system should treat this as a completely new and independent task, with no contextual connection to previous original or translated text (e.g., it should not automatically refer to the subject of the previous sentence).
- 2. Each translation result should be displayed independently on the right side, without stacking historical translations (historical records are saved separately in the left-side list).

3.3.2 File Translation

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3.4 Results Display and Download

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3.5 Historical Record Management

3.6 Interface and Experience

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3.3 Non-functional Objectives

Dimension | Objective

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Performance | Text translation response ≤ 2 seconds; File translation response ≤ 30 seconds (for documents up to 10 pages)

Availability | System annual availability $\geq 99.5\%$

Security | Data encrypted at rest and in transit using HTTPS; user sessions isolated

Compatibility | Supports latest two versions of Chrome and Edge; resolution 1280×720 or higher

Maintainability | Terminology database can be independently updated without requiring a release deployment